

Gradient Boosting Regression – Mathematics

1. Algorithm

Gradient Boosting is an **additive model**:

$$F_M(x) = \sum_{m=1}^M \gamma_m h_m(x)$$

Where:

- $F_M(x)$ → final model after M iterations
- $h_m(x)$ → weak learner (usually a regression tree)
- γ_m → weight (step size) for that learner

👉 Each new tree is added to the ensemble, improving the fit step by step.

2. Additive Modelling

- Instead of learning one big model at once, Gradient Boosting builds the model **additively** (tree by tree).
- At iteration m :

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$$

- The new tree $h_m(x)$ is trained to fix the errors made by $F_{m-1}(x)$
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3. Gradient Descent View

We want to minimize a loss function $L(y, F(x))$.

At each step:

1. Compute **negative gradient** (residuals):

$$r_{im} = - \left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \right]_{F(x)=F_{m-1}(x)}$$

👉 These residuals represent the direction of steepest descent.

2. Fit a weak learner $h_m(x)$ to predict these residuals.
3. Find the optimal multiplier γ_m :

$$\gamma_m = \arg \min_{\gamma} \sum_i L(y_i, F_{m-1}(x_i) + \gamma h_m(x_i))$$

4. Update the model:

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$$

4. Runge's Phenomenon

- When using **high-degree polynomials** in interpolation, the function can oscillate wildly (overfitting).
- Gradient Boosting avoids this problem by building the model **incrementally** with many small, simple learners (trees), rather than fitting a complex function all at once.

5. Visualization

- Imagine a curve-fitting process where:
 1. First, a simple function tries to fit the data.
 2. Residuals are computed.
 3. A new function is added to fit those residuals.
 4. Repeated many times → the final curve gets closer to the actual data smoothly.

6. Kaggle Demo

- Shows Gradient Boosting applied practically on a dataset.
 - You'll see how each tree reduces error slightly.
 - With enough iterations, the model becomes very accurate while controlling variance using **learning rate** and **tree depth**.
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7. Explanation

- Gradient Boosting = **Additive + Gradient Descent optimization.**
- Instead of solving directly, it improves step by step.
- Very flexible since you can define different loss functions:
 1. MSE (for regression)
 2. Log loss (for classification)
 3. Even custom loss functions.

➤ Summary in Simple Words:

Gradient Boosting Regression builds the model step by step, each new tree correcting the residuals (errors) of the previous one. Mathematically, it's an **additive model** optimized with **gradient descent** on the chosen loss function. This makes it powerful, flexible, and stable against overfitting compared to fitting one complex model.
